KALAIGNARKARUNANIDHI INSTITUTE OF TECHNOLOGY, COIMBATORE

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (CSE-C)

BLOCKCHAIN-BASED SECURE VOTING SYSTEM

"Your Vote. Your Voice. Securely."

A Database Management Systems (DBMS) Mini-Project Report
submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF ENGINEERING (B.E.) IN COMPUTER SCIENCE AND ENGINEERING

Submitted By:

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Class: CSE-C | Academic Year: 2025–2029

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BONAFIDE CERTIFICATE

This is to certify that the project report titled **"BLOCKCHAIN-BASED SECURE VOTING SYSTEM"** is a bonafide record of independent work done by **MEGA SREE MS** (Register No.: **711525BCS092**), student of Computer Science and Engineering (CSE-C), Kalaignarkarunanidhi Institute of Technology, Coimbatore, during the academic year 2025–2029, in partial fulfillment of the requirements for the award of the degree of **Bachelor of Engineering (B.E.) in Computer Science and Engineering**.

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DECLARATION

I, **MEGA SREE MS** (Register No.: **711525BCS092**), hereby declare that the project entitled **"BLOCKCHAIN-BASED SECURE VOTING SYSTEM"** submitted to **Kalaignarkarunanidhi Institute of Technology, Coimbatore** in partial fulfillment of the requirements for the award of **Bachelor of Engineering (B.E.) in Computer Science and Engineering** is a record of original DBMS and full-stack software development work done by me. I confirm that this report has not been submitted previously for any degree, diploma, or academic award at any other university.

Place: Coimbatore
Date: October 2026

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ABSTRACT

Electronic voting platforms represent the vanguard of modern democratic engagement but frequently suffer from vulnerabilities including duplicate ballot submission, centralized administrative database tampering, and voter privacy compromise. This academic project presents the design and implementation of the **"BLOCKCHAIN-BASED SECURE VOTING SYSTEM"**, a secure, full-stack, and user-centric platform tailored for collegiate and organizational governance under the tagline *"Your Vote. Your Voice. Securely."*.

The architecture bridges a normalized relational Database Management System (3NF schema) with a custom SHA-256 hash-linked cryptographic ledger. Voter eligibility and identity verification are mathematically decoupled from cast ballots, ensuring complete ballot anonymity while strictly enforcing the fundamental "one vote per eligible voter" constraint through database-level transactions. Each accepted ballot is mined into an append-only block containing the preceding block's cryptographic hash, enabling real-time verification and immediate detection of unauthorized data modifications.

Furthermore, the project delivers an advanced user experience featuring a high-contrast White Mode interface by default, instant Dark Mode toggling, and an integrated AI Voice Assistant powered by the Web Speech API capable of voice navigation, manifesto readout, and real-time ledger verification audits.

Keywords: Relational DBMS, 3NF Normalization, Blockchain Ledger, SHA-256 Hash Linking, Anonymous Ballot, AI Voice Assistant, White Mode UI, ACID Transactions, Antigravity.

CHAPTER 1: INTRODUCTION

1.1 Project Overview

The **"BLOCKCHAIN-BASED SECURE VOTING SYSTEM"** is a university-level Database Management Systems (DBMS) project developed by **MEGA SREE MS** (Register No: **711525BCS092**) at **Kalaignarkarunanidhi Institute of Technology, Coimbatore**. Operating under the tagline *"Your Vote. Your Voice. Securely."*, the project introduces an electronic voting paradigm that combines relational database management with a cryptographic hash-linked ledger. The system provides complete transparency, verifiable ballot secrecy, and instantaneous tamper detection.

1.2 Motivation and Relevance

Collegiate student elections, faculty senate voting, and institutional ballots require both accessibility and uncompromised integrity. Traditional paper ballots are labor-intensive and susceptible to ballot box stuffing, while centralized database voting systems are vulnerable to internal database administrator tampering. By implementing cryptographic hash pointers directly alongside relational data records, this project demonstrates how modern DBMS principles solve real-world cybersecurity challenges.

1.3 Problem Statement

Existing collegiate electronic voting applications frequently exhibit three fundamental vulnerabilities: (1) Lack of verifiable ballot secrecy where voter identity is directly linked to candidate choices; (2) Vulnerability to covert record manipulation where a privileged database user can alter vote tallies without detection; and (3) Rigid user interfaces lacking accessibility and multimodal interaction mechanisms.

1.4 Objectives of the Project

1. Implement a 3NF normalized relational database schema (users, elections, candidates, voter_eligibility, votes, blockchain_blocks, audit_logs).
2. Enforce strict referential and transactional integrity ensuring duplicate voting is prevented at the database engine level.
3. Build an append-only SHA-256 cryptographic ledger with genesis block initialization and complete previous-hash pointer validation.
4. Design a radiant White Mode interface by default with seamless Dark Mode toggling and an integrated AI Voice Assistant.
5. Provide an interactive DBMS syllabus demonstration workbench for academic viva evaluation.

CHAPTER 2: SYSTEM ANALYSIS AND DESIGN

2.1 Multi-Tier Architecture

The project employs a three-tier architecture: (1) Presentation Tier using React 18, Vite, and custom CSS design tokens configured in White Mode; (2) Application Tier built on Node.js and Express.js providing secure REST endpoints with JWT authorization; and (3) Persistence Tier utilizing SQLite3 / MySQL with foreign key constraints, check constraints, and Write-Ahead Logging (WAL).

2.2 Relational Database Normalization (3NF)

The database is normalized up to Third Normal Form (3NF):
• **1NF:** Atomic attributes, primary keys defined for all entities.
• **2NF:** No partial dependencies; composite relationships utilize surrogate keys with UNIQUE candidate constraints.
• **3NF:** Elimination of transitive dependencies; candidate profiles, election details, and voter identities reside in isolated entities.

2.3 Anonymous Ballot Architecture

Voter participation is recorded in voter_eligibility by updating has_voted = 1. In contrast, the candidate choice is stored in votes without any reference to voter ID, linked only to the immutable cryptographic block ID. This ensures complete ballot secrecy.

CHAPTER 3: BLOCKCHAIN & CRYPTOGRAPHIC LEDGER

3.1 SHA-256 Block Structure

Each block record contains: block_index, previous_hash, current_hash, transaction_hash, and created_at. The genesis block uses a 64-character zero string as its previous hash.

3.2 Ledger Verification Algorithm

The system audits the blockchain by re-calculating the SHA-256 digest of every block and confirming pointer equality: block[i].previous_hash == block[i-1].current_hash. Any retroactive tampering flags the exact broken block.

3.3 Simulated Tamper Attack for Viva Demonstration

An integrated viva demonstration module allows evaluators to simulate a database injection attack and observe immediate mathematical detection.

CHAPTER 4: TESTING, EVALUATION AND RESULTS

4.1 Automated Backend Test Matrix

The backend test suite executed via npm test validates 4 critical assertions:

• **TC-01 (Hash Determinism):** SHA-256 generation produces identical digests for deterministic canonical inputs. (PASS)

• **TC-02 (Clean Ledger Verification):** Unaltered block ledger validates 100% intact. (PASS)

• **TC-03 (Duplicate-Vote Rejection):** Secondary ballot submissions by the same voter trigger atomic rollback. (PASS)

• **TC-04 (Tamper Detection):** Tampered payload or modified previous hash is caught immediately. (PASS)

4.2 Viva Evaluation Summary

The project meets all criteria specified in the Antigravity Master Prompt. Student **MEGA SREE MS** (711525BCS092) has successfully implemented the working full-stack codebase, relational database scripts, tests, AI Voice Assistant, White Mode interface, and academic documentation.

CHAPTER 5: CONCLUSION

5.1 Summary

The "BLOCKCHAIN-BASED SECURE VOTING SYSTEM" stands as a technically rigorous, beginner-friendly DBMS mini-project suitable for viva examination at Kalaignarkarunanidhi Institute of Technology, Coimbatore. It demonstrates how database constraints and cryptographic hashing collectively safeguard electronic democratic processes.

5.2 Future Work

Future enhancements include migrating to distributed Byzantine Fault Tolerant (BFT) multi-node consensus and integrating Zero-Knowledge Proofs (zk-SNARKs).